

CSE 321 - Introduction to Algorithm Design

HOMEWORK 01

① Prove the statements below whether its true or not

Ⓐ It is true for all $f(n)$ and $g(n)$ functions $f(n) \in O(g(n))$ or $g(n) \in O(f(n))$.

$$f(n) \leq c_1 \cdot g(n) \quad \text{veya} \quad g(n) \leq c_2 \cdot f(n)$$

Asitliğin sağlanıyor olması gerektiği fakat burada

"=" ile durumu set kavuşu olduğu için constant değerlere bağlı bir değişim olması sonucu etkiler.

Tek bir örnek bu durumu olmadığını göstereceğinden bir örnek vererek açıklayalım.

$$f(n) = n^2 \quad g(n) = n^3 \quad \text{olsun bu durumda}$$

$f(n) \in O(g(n))$ durumu sağlanır

$$n^2 \leq c \cdot n^3 \quad \checkmark$$

fakat tersi set kavuşu olmaz

$$f(n) \notin O(g(n))$$

$$\underline{\underline{n^2 \notin c \cdot n^3 \text{ 'tür.}}}$$

(2)

(b) Let $f(n)$, $g(n)$ and $h(n)$ be functions. If $f(n) \in O(g(n))$ then the following is true:

$$\frac{f(n)}{h(n)} \in O\left(\frac{g(n)}{h(n)}\right)$$

$$f(n) \in O(g(n)) \\ = f(n) \leq c \cdot g(n)$$

$$\hookrightarrow \frac{f(n)}{h(n)} \stackrel{?}{\in} O\left(\frac{g(n)}{h(n)}\right)$$

$$\frac{f(n)}{h(n)} \leq c \cdot \frac{g(n)}{h(n)}$$

$h(n)$ = pozitif katsayılı bir fonksiyon ise;

$$\frac{f(n)}{h(n)} \leq c \cdot \frac{g(n)}{h(n)} \text{ durumu her sort için sağlanır}$$

$h(n)$ = negatif katsayılı bir fonksiyon ise;

$$\frac{f(n)}{h(n)} \geq c \cdot \frac{g(n)}{h(n)} \Rightarrow \frac{f(n)}{h(n)} \in \Omega\left(\frac{g(n)}{h(n)}\right)$$

durumu ise doğrudur

$\Rightarrow$ Eşitlik her durum için sağlanmaz

(c) Let $f(n)$ and $g(n)$ be functions and k be an integer. If $f(n) \in O(g(n))$ then the following is true: $f(n)^k \in O(g(n)^k)$

$$f(n) \leq c \cdot g(n) \stackrel{?}{\Rightarrow} f(n)^k \stackrel{?}{\in} O(g(n)^k)$$

$$f(n)^k \leq c \cdot g(n)^k$$

$k \geq 0$ ise $f(n)^k \leq c \cdot g(n)^k$ durumu sağlanır.

$k < 0$ ise $f(n)^k \geq c \cdot g(n)^k$ durumu gerçektir.

Bundan dolayı yukarıdaki durum her koşulda doğru değildir.

② Prove or disprove the following statements.

① $n^3 \in o(2^n)$

Limit yöntemi ile Gözümlü;

$$\lim_{n \rightarrow \infty} \frac{n^3}{2^n} = \lim_{n \rightarrow \infty} \frac{3n^2}{1 \cdot \ln^2 2^n} = \lim_{n \rightarrow \infty} \frac{6n}{(\ln 2)^2 2^n} = \lim_{n \rightarrow \infty} \frac{6}{(\ln 2)^3 2^n}$$

↑
L'Hospital

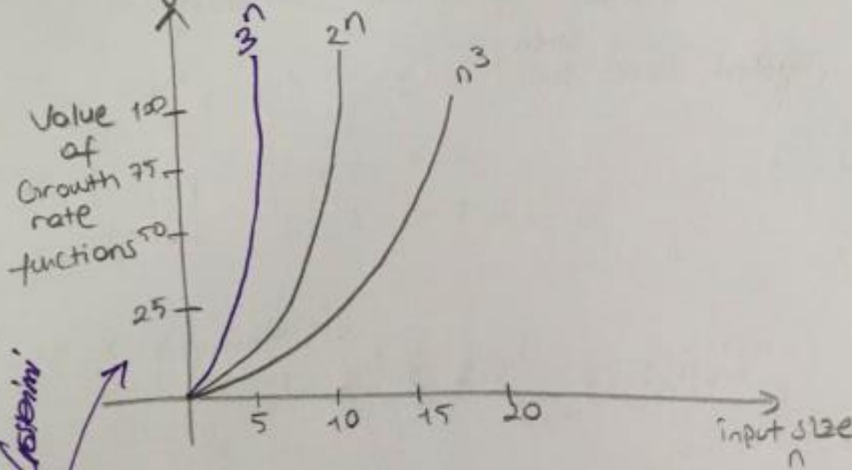
↑
Tercih
L'Hospital
kullanılır
uygunlarsak

$$= \frac{1}{\infty} = 0$$

$$n^3 \in o(2^n)$$

PROVE

Büyüme Grafiklerini Göstermek



② $2^n \in o(3^n)$

$2^n < c \cdot 3^n$ koşulu sağlanması yeterlidir Little o özelliğini sağlayabilmek için limit sonucunun 0 olması gerekir.

$$\lim_{n \rightarrow \infty} \frac{2^n}{3^n} = \lim_{n \rightarrow \infty} \frac{\ln 2 \cdot 2^n}{\ln 3 \cdot 3^n}$$

$$n > 0$$

$$n > n_0$$

$$\lim_{n \rightarrow \infty} \frac{(\ln 2)^2 \cdot 2^n}{(\ln 3)^2 \cdot 3^n} = \frac{\text{Sayı}}{\text{Sonsuz}} = 0 \text{ sonucuna ulaşacağız}$$

Induction ile

$c=1 \Rightarrow$
 $2^1 \leq 1 \cdot 3^1$

$n=k$ için $\Rightarrow 2^k \leq 1 \cdot 3^k$

$n=k+1$ için $\Rightarrow 2^{k+1} \leq 3^{k+1}$

Gösterebiliriz $2^k \cdot 2 \leq 3^k \cdot 3$ ✓

Tüm durumlar için sağlanmıştır

$$\textcircled{3} \quad c - n! \in o(100^n)$$

$$\text{Stirlings Formula: } n! \cong \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

$$\lim_{n \rightarrow \infty} \frac{n!}{100^n} = \lim_{n \rightarrow \infty} \sqrt{2\pi n} \left(\frac{n}{100e}\right)^n = \infty$$

$$\Rightarrow \underline{\underline{n! \in \omega(100^n)}}$$

- ③ Write the pseudocode of linear search with repeated elements.
Analyze its best case, worst case and average case complexities.

Pseudo-Code:

```

function LinearSearch( L[1:n], x )
  for i = 1 to n do
    if ( L[i] = x ) then
      return (i)
    end if
  end for
  return  $\phi$ 
end

```

↳ Listede tekrarlı elemanlardan verilen elemana ait geçtiği ilk konumunu bulacak şekilde best, worst, average case hesaplanmıştır.

Best Case: $x = L[1] \Rightarrow$ best case : $B(n) \in \Theta(1)$ 'dir.

Worst case: $x = L[n]$ veya listede hiç yoksa;
 $w(n) \in \Theta(n)$

Average Case:

$$E(i) = \sum_{i=1}^{w(n)} i \cdot P(i)$$

↑ i 'nin bulunma olasılığı
Prob. that x occurs in the i th position

Eleman listede kaç kez geçti

$$P_i \cdot \frac{x}{n} = \frac{P_x}{n}$$

$$\frac{P_x}{n} + (1-p)$$

↑ listede olması durumu

$$\begin{aligned}
 & \sum_{i=1}^{n-1} i \frac{P_x}{n} + n \left(\frac{P_x}{n} + (1-p) \right) \\
 &= \frac{P_x}{n} \frac{n(n-1)}{2} + \frac{P_x + n(1-p)}{1} \\
 &= \frac{P_x(n-1) + 2P_x + 2n(1-p)}{2} \\
 &= \frac{P_x n + P_x + 2n - 2np}{2} \\
 &= \frac{P_x(n+1) + 2n(1-p)}{2} \in \underline{\underline{\Theta(n)}}
 \end{aligned}$$

④ Enigma algorithm is given below.

Algorithm Enigma ($A[0 \dots n-1, 0 \dots n-1]$)

// input the matrix A of real numbers

for $i=0$ to $n-2$ do

for $j=i+1$ to $n-1$ do

if $A[i,j] \neq A[j,i]$

return false

return true

a) What is problem size (input size)?

$(n-1)(n-1)$ = input size

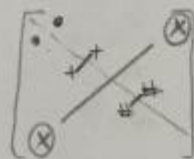
b) What is the main operation of the algorithm?

Compare Operation

if $A[i,j] \neq A[j,i]$

c) What is calculated by this algorithm?

Simetrik matris olup olmadigi kontrol edilmistir.



→ Bu sekilde elemanlar karilastirilir.

d) Show this algorithm's complexity using the O , Ω and Θ asymptotic notations for following states below

i. Worst Case Complexity:

$$W(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} [(n-1) - (i+1) + 1]$$

$$= \sum_{i=0}^{n-2} (n-1-i) = (n-1) + (n-2) + \dots + 1 = \frac{n(n-1)}{2}$$

= Quadratic function

$$W(n) \in \Theta(n^2)$$

ii. Best Case Complexity:

$A[0,1]$ & $A[1,0]$ elemaninin farkli olmasi durumunda biter.

$$\underline{\underline{B(n) \in O(1)}}$$

5) You are given the following function.

function myFun2(A[1...n])

if ($n \leq 1$)

return 4

else

return myFun2(A[1...n/3]) + myFun2(A[2n/3...n])

a) What is problem size?

$n \rightarrow$ Arrayın eleman sayısı

$n = 3^k$ olsun

$$C(3^k) = 2 C(3^{k-1}) + 1$$

Substitution
method

$$= 2 [2 C(3^{k-2}) + 1] + 1$$

$$= 2^2 C(3^{k-2}) + 3 + 1$$

$$= 2^i C(3^{k-i}) + 3^{i-1} + 3^{i-2} + \dots + 1$$

$$= 2^k (C(3^{k-k})) + 3^{k-1} + \dots + 1$$

$$= \underline{\underline{3^k - 1}}$$

$$= \underline{\underline{n-1}} \quad \underline{\underline{\Omega(n)}}$$

Main Operation Compare

⑥ Let A be a set, n be an even number, and A consist of n positive integers. Design an efficient algorithm to separate set A into set A_1 and set A_2 , each of which includes $n/2$ items, in order to maximize the difference between the addition of elements in set A_1 and the addition of elements in set A_2 . Analyze and show your algorithm's complexity using proper asymptotic notations.

procedure maximizeDifference ($A[0 \dots n]$)

if $n \geq 1$

THEN $q = \text{FLOOR}(n/2)$

MERGE ($A, 0, q$)

MERGE ($A, q+1, r$)

MERGE ($A, 0, q, n$)

end if

for $i = 0$ to $\lfloor n/2 \rfloor$

set $A_1 += A[i]$

end for

for $i = \lfloor n/2 \rfloor + 1$ to n

set $A_2 += A[i]$

end for

result = set A_2 - set A_1

end

MERGE (A, p, q, r)

$n_1 \leftarrow q - p + 1$

$n_2 \leftarrow r - q$

Create arrays $L[1 \dots n_1+1]$ and $R[1 \dots n_2+1]$

FOR $i \leftarrow 1$ TO n_1

DO $L[i] \leftarrow A[p+i-1]$

FOR $j \leftarrow 1$ TO n_2

DO $R[j] \leftarrow A[q+j]$

$L[n_1+1] \leftarrow \infty$

$R[n_2+1] \leftarrow \infty$

$i \leftarrow 1$

$j \leftarrow 1$

FOR $k \leftarrow p$ TO r

DO IF $L[i] \leq R[j]$

THEN $A[k] \leftarrow L[i]$

$i \leftarrow i + 1$

ELSE $A[k] \leftarrow R[j]$

$j \leftarrow j + 1$

← Set A_1
← Set A_2
} merge'ler $\Theta(n \log n)$

} $\Theta(n)$ kadar oradaolu
elementlerin
eder

Toplamda $\Theta(n \log n)$
kadar istenide algoritma
sonucuna erisilebilir.

- ⑦ Design a recursive algorithm to find out how many 0s are included in an array which consist of only 0s and 1s. Analyze and show your algorithm's complexity using proper asymptotic notations.

Öncelikle quick sort kullanarak arrayi sıralayıp sonrasında binary search yardımıyla 0'ların sayısını elde edilecek

procedure find0s(A[], n)

quickSort(A, 0, n) $\rightarrow \Theta(n \log n)$

var count

for i = 0 to n

if A[i] == 0

count++

end if

end for

return count

end

procedure quickSort(A, 0, n)

var q

if n > 1

q = partition(A, p, r)

quickSort(A, p, q-1)

quickSort(A, q+1, r)

end if

end for

procedure partition(A, p, r)

var tmp

var x = A[r]

var i = p-1

for j = p to r-1

if A[j] ≤ x

++i

tmp = A[r]

A[r] = A[j]

A[j] = tmp

end if

end for

tmp = A[r+1]

A[r+1] = A[r]

A[r] = tmp

return i+1

Cresel algoritma $\Theta(n \log n)$ 'de çalışacak

$\Theta(n \log n)$

iki ayrı kola ayrı recursive şekilde sıralama yapacak $\Theta(n \log n)$

⑧ Solve the recursive equations below. Calculate exact values and explain using theta notation.

① $T(n) = -4T(n-1) - 4T(n-2)$ $T(0)=0$ $T(1)=1$

↳ Homogeneous denklemin içinde yazarsak;

$$r^2 + 4r + 4 = 0 \quad (r+2)^2 \Rightarrow \underline{\underline{r = -2}}$$

$$\alpha_n = \alpha_1 (-2)^n + \alpha_2 n(-2)^n$$

$$\alpha_0 = 0 = \alpha_1$$

$$\alpha_1 = 1 = \alpha_1 (-2) + \alpha_2 (-2) \Rightarrow \alpha_1 = 0$$

$$\alpha_n = n(-2)^n - \frac{1}{2} \quad \underline{\underline{\alpha_2 = -\frac{1}{2}}}$$

$$\underline{\underline{\Theta(2^n)}}$$

② $T(n) = T(n-1) + 6T(n-2)$ $T(0)=3$ $T(1)=6$

$$r^2 - r - 6 = 0 \quad r = 3 \quad r = -2$$

$$\alpha_n = \alpha_1 3^n + \alpha_2 (-2)^n$$

$$\alpha_0 = 3 = \alpha_1 + \alpha_2$$

$$\alpha_1 = 6 = \alpha_1 3 + \alpha_2 (-2)$$

$$\alpha_1 = \frac{12}{5}$$

$$\alpha_2 = \frac{3}{5}$$

$$\alpha_n = \frac{12}{5} 3^n + \frac{3}{5} (-2)^n$$

$$\underline{\underline{\Theta(3^n)}}$$

③ $T(n) = -5T(n-1) - 6T(n-2) + 42 \cdot 4^n$ $T(1)=56$
 $T(2)=278$

$$r^2 + 5r + 6 = 0$$

$$\alpha_n = \alpha_1 (-3)^n + \alpha_2 (-2)^n$$

$$C \cdot 42 \cdot 4^n = -5C \cdot 42 \cdot 4^{n-1} - 6C \cdot 4^{n-2} + 42 \cdot 4^n$$

$n=2$ için $42 \cdot 16 C = -42 \cdot 5 \cdot 4C - 6C + 42 \cdot 16$

$$C = \frac{42 \cdot 16}{42 \cdot 26 + 6}$$

$$\underline{\underline{\Theta(4^n)}}$$

(9) Give exact solutions for $T(n)$ in each of the following recurrences.

$$\begin{aligned}
 \textcircled{a} \quad T(n) &= T(n-1) + (n^2 + 1) \quad T(0) = 3 \\
 &= [T(n-2) + (n-1)^2 + 1] + n^2 + 1 \\
 &= [T(n-3) + (n-2)^2 + 1] + (n-1)^2 + 1 + n^2 + 1 \\
 &= [T(n-4) + (n-3)^2 + 1] + (n-2)^2 + 1 + (n-1)^2 + 1 + n^2 + 1 \\
 &\vdots \\
 &= T[n-k] + \sum_{i=1}^k (n-i)^2 + i \\
 &= T(0) + \frac{n(n-1)(2n-1)}{6} + \frac{n(n-1)}{2} \\
 &= \underset{(6)}{3} + \frac{2n^3 - n^2 - 2n^2 + n}{6} + \underset{(3)}{\frac{n^2 - n}{2}} \\
 &= \frac{18 + 2n^3 - \cancel{2n^2} + n + \cancel{3n^2} - 3n}{6} \Rightarrow \frac{18 + 2n^3 - 2n}{6} \quad //
 \end{aligned}$$

$$\textcircled{b} \quad a_n = \sum_{k=1}^n k + n^2$$

$$a_n = a_{n-1} + n + n^2$$

$$a_n = a_n \quad \Rightarrow \quad a_n = C(1)^n$$

$$\hookrightarrow a_n = \frac{n^2}{2} + \frac{n}{2} + n^2 \Rightarrow \frac{n(n+1)}{2} + n^2$$

$$\Rightarrow \frac{2n^2 + n}{2} \quad \theta(n^2)$$

10) Solve the recursive equations below asymptotically.

a) $f(n) = 3f(n/2) + n^2$ $f(1) = 1$

Bu sorular "master Theorem" kullanılarak yapılacaktır.

$$T(n) = aT(n/b) + f(n)$$

$a \geq 1$, $b > 1$ are constants, $f(n)$ is an asymptotically positive function

I - If $f(n) = O(n^{\log_b a - \epsilon})$ for some constants $\epsilon > 0$, then $T(n) = O(n^{\log_b a})$

II - If $f(n) = \Theta(n^{\log_b a} \log^k n)$ with $k \geq 0$, then $T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$

III - If $f(n) = \Omega(n^{\log_b a + \epsilon})$ with $\epsilon > 0$, and $f(n)$ satisfies the regularity condition, then $T(n) = \Theta(f(n))$ regularity condition:

$a f(n/b) \leq c f(n)$ for some constant $c < 1$ and all sufficiently large n .

$\frac{a=3}{b=2}$ $f(n) = n^2$ $O(n^{\log_2 3 - \epsilon})$

$\uparrow$ n^2 'den küçük bir değer oluşturunca sonucu ulaşıyoruz

$\Theta(n^{\log_2 3} \log^k n)$

$\uparrow$ n^2 'den küçük değeri sağlayamıyoruz

$\Omega(n^{\log_2 3 + \epsilon})$

$\uparrow$ değeri sağlayabilmeli bir değere sahip

$\log_2 4 \rightarrow 2$ $\log_2 3 + 2 \checkmark$

$\Theta(T(n) = \Theta(n^2)) \leftarrow \text{CASE 3}$

b) $f(n) = 3f(n/2) + n^2 \log n$ $f(1) = 1$

$\frac{a=3}{b=2}$

3. case 'i sağlayacak

$\Omega(n^{\log_2 3 + \epsilon})$

$\Rightarrow T(n) = \Theta(n^2 \log n)$